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BME 105: Genetics in the Genomics Era
Spring 2019: Final

Name: _____

Student ID:_____

Before starting, write your initials on the top of each page.

You can use the back-side of the pages for scratch, but we will not grade answers written on the back-side of the page.

Section	**Points**
Part I: One- or two-word answers, True/False, matching	/6
Part II: Multiple choice	/38
Part III: Short answer	/46
Part IV: Extended calculation	/60
Total Points	**/150**

Second Letter

1st letter	U		C		A		G		3rd letter
U	UUU UUC UUA UUG	Phe Leu	UCU UCC UCA UCG	Ser	UAU UAC UAA UAG	Tyr Stop Stop	UGU UGC UGA UGG	Cys Stop Trp	U C A G
C	CUU CUC CUA CUG	Leu	CCU CCC CCA CCG	Pro	CAU CAC CAA CAG	His Gln	CGU CGC CGA CGG	Arg	U C A G
A	AUU AUC AUA AUG	Ile Met	ACU ACC ACA ACG	Thr	AAU AAC AAA AAG	Asn Lys	AGU AGC AGA AGG	Ser Arg	U C A G
G	GUU GUC GUA GUG	Val	GCU GCC GCA GCG	Ala	GAU GAC GAA GAG	Asp Glu	GGU GGC GGA GGG	Gly	U C A G

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Part I: One- or two-word answers, T/F, and matching. Each question is worth 2 points.

1. _____gene pool_____ sum total of all alleles carried in all members of a population

2. __genotype frequency_____ proportion of individuals in a population that are of a particular genotype

3. _____gene drive_____ a strategy that uses CRISPR-Cas9 to rapidly introduce a genetic trait across an interbreeding population

Part II: Multiple choice (circle your answer. If there is more than one correct answer, circle all correct answers, unless otherwise specified) Each question is worth 2 points.

1. Suppose that in plants, smooth seeds (*S*) is dominant to wrinkled seeds (*s*) and tall plants (*T*) is dominant to short plants (*t*). An SsTt plant was crossed to a plant that was short and wrinkled. What proportion of the progeny is expected to be heterozygous for tall and smooth?

   a. 1/2

   b. 1/4

   c. 1/8

   d. 1/16

   e. 0

   b

2. You want to clone and express the cDNA copy of a human gene, named Gene Z in E.coli bacterial cells. The following is a schematic of Gene Z cDNA that is already cloned in a different vector. Only the cDNA and flanking regions are shown. The recognition sites of different restriction enzymes are shown.

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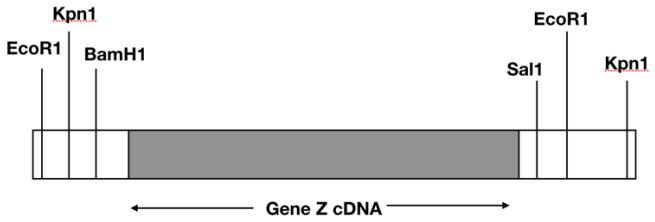
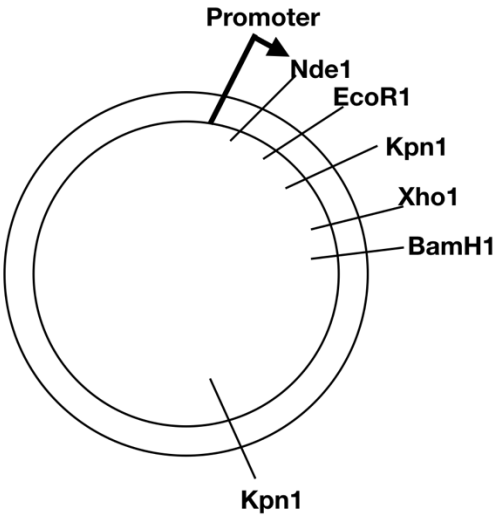


BamH1	Nde1
5′G/GATCC3′	5′CA/TATG3′
3′CCTAG/G5′	3′GTAT/AC5′
EcoR1	**Kpn1**
5′G/AATTC3′	5′GGTAC/C3′
3′CTTAA/G5′	3′C/CATGG5′
Sal1	**Xho1**
5′G/TCGAC3′	5′C/TCGAG3′
3′CAGCT/G5′	3′GAGCT/C5′

You want to use the following plasmid as the vector for cloning Gene Z cDNA. Which of the following strategies would be the best to get expression of Gene Z cDNA? Circle only one choice.



Strategy	Restriction enzymes used for Gene Z cDNA	Restriction enzyme used for vector
a	EcoR1	EcoR1
b	EcoR1	Nde1 & BamH1
c	EcoR1 & Sal1	EcoR1 & Xho1
d	BamH1 & Sal1	BamH1 & Xho1

C

3. How was the cDNA generated?
   a. Designing a forward primer for the first exon and a reverse primer of the last exon and amplifying from genomic sequence

   b. Designing a forward primer for the last exon and a reverse primer of the first exon and amplifying from genomic sequence

   c. Identify restriction sites from the genomic sequence flanking the gene

   d. Convert the Gene Z mRNA to cDNA then design a forward primer for the first exon and reverse primer for the last exon and amplify the sequence.

   d

4. What aspect of chromosome behavior most clearly accounts for Mendel's law of independent assortment?
   a. movement of sister chromatids to opposite poles at anaphase II of meiosis

b. movement of homologous chromosomes to opposite poles at anaphase I of meiosis

c. crossing over between homologous chromosomes during prophase I of meiosis

d. replication of chromosomes prior to meiosis

e. independent alignment of different homologous pairs on the metaphase I spindle

e

5. Which of the following is one of the assumptions of the Hardy-Weinberg law?
   a. The population is small.

   b. Non-random mating occurs within the population.

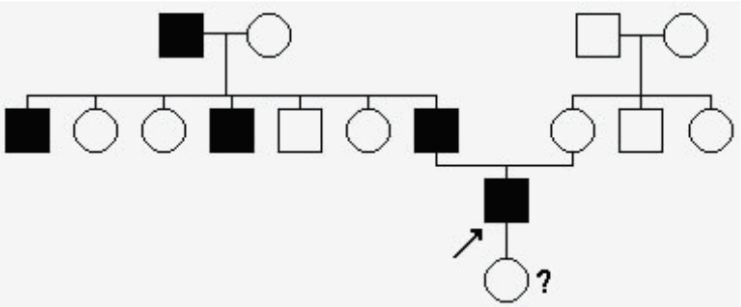
   c. Mutations appear in the gene pool at high frequency.

   d. Migration occurs into or out of the population.

   e. The ability of all genotypes to survive and reproduce is the same.

e

6. A trait has a population frequency of 1 in 40,000. Assuming the most likely mode of inheritance as shown in the pedigree, what is the probability that the proband's daughter has inherited the familial mutation?



a. 0

b. ¼

c. ½

d. 1

c

7. Normal fruit flies have red eyes. Flies with mutations that block the function of the X-linked *white* gene have white eyes. The wild-type allele is completely dominant to mutant alleles. A cross between a white-eyed female fruit fly and a red-eyed male would generate which of the following types of progeny?

   a. red-eyed females and white-eyed males

   b. white-eyed females and red-eyed males

   c. all red-eyed females and a 50:50 mixture of white-eyed and red-eyed males

   d. all red-eyed males and a 50:50 mixture of white-eyed and red-eyed females

   e. The result cannot be predicted because it depends on whether the female is homozygous or heterozygous.

   a

8. Transgenic animals made by pronuclear injection can be limited for studies of the links between genes and disease because

   a. the transgene allele must be dominant to the normal, endogenous allele and some diseases are caused by recessive mutations.

   b. the transgene allele must be recessive to the normal, endogenous allele and some diseases are caused by dominant mutations.

   c. all diseases are caused by mutations in multiple genes.

   d. diseases are not caused by specific alleles of a gene, but by posttranslational modifications of proteins.

   a

9. Most scientists believe that genome editing in the germline of humans should not currently be performed. Which of the following are possible reasons for this?
    a. It is not technically feasible

    b. Most complex traits involve variants at multiple loci

    c. More research is necessary to understand the potential risks

    d. More research is necessary to make genome editing more precise and reduce unintended off-target editing.

    b, c, d

10. What is the most common reason that new mutations are lost from a population?
    a. Most mutations are caused by exposure to environmental mutagens that result in many mutations in the same individual, leading to death.

    b. Most mutations are silent.

    c. All new mutations result in lower reproductive fitness.

    d. Populations are of a finite size and new mutations can be lost by chance.

    d

11. Most complex quantitative traits
    a. are affected by both genetic and environmental factors.

    b. usually do not result in continuous phenotypes.

    c. are not usually affected by environmental conditions.

    d. usually result in one or two phenotypic values such as red or white eye color in *Drosophila*.

    a

12. A scientist wants to determine whether a mutant phenotype is due to the loss of gene *C* or gene *D*, which are both deleted in a mutant fly with abnormal eyes. Mutant flies with transgenic gene *C* have wild-type eyes, whereas mutant flies with transgenic gene *D* have abnormal eyes. What can be concluded from these results?
    a. Either gene *C* or gene *D* is required for normal eye development.

b. Both gene *C* and gene *D* are required for normal eye development.

c. Gene *C* is required for normal eye development.

d. Gene *D* is required for normal eye development.

c

13. A researcher performs RNAi against a gene that is required for viability in *C. elegans* embryos and 70% of the embryos die during development. Some of the embryos may have survived because
    a. RNAi did not eliminate the mRNA (and protein) of the gene completely.

    b. the dsRNA was used to translate additional proteins to compensate for the mRNA that was degraded.

    c. RNAi eliminated the protein product of the gene.

    d. of issues related to fertilization.

    a

14. To create a knockout mouse, after introducing to ES cells a DNA construct in which a specific gene is mutagenized by the insertion of a drug resistance marker, the cells
    a. that have incorporated the transgene by homologous recombination grow in the presence of the drug and these are injected into a host blastocyst.

    b. that lack the transgene grow in the presence of the drug and these are injected into a host blastocyst.

    c. that have deleted the transgene grow in the presence of the drug and the nuclei are injected into a host blastocyst.

    d. that lack the transgene grow in the presence of the drug and the nuclei are used for nuclear transfer.

    a

15. In primroses, the dominant allele of gene *K* is necessary to synthesize blue flower pigment. Blue pigment synthesis is inhibited by a dominant allele of gene *D*. In other words, plants with the genotype *K*- *D*- will not produce pigment (and their flowers will be white) because of the presence of the *D* allele.

If two dihybrid plants (*Kk Dd*) are crossed, what is the ratio of blue to white offspring in the progeny?

a. 3 blue: 1 white

b. 7 blue: 9 white

c. 4 blue: 12 white

d. 3 blue: 13 white

e. 1 blue: 15 white

d

16. What type of gene interaction determines flower color in primroses?

a. one gene with two completely dominant alleles

b. two-genes with recessive epistasis

c. two redundant genes

d. two genes with dominant epistasis

d

17. Suppose that researchers identified two different regions of the genome with statistically significant associations of SNPs with a certain disease. In Region 1, SNPs extending over 200 kb (200,000 bases) of DNA showed such associations. In Region 2, the region containing disease-associated SNPs was 2 Mb (2,000,000 bases) long. If you assume that each of these two regions has only one disease-causing variant, which of the two mutations was likely to have occurred earlier in human history?

a. Region 1

b. Region 2

a

18. Sickle cell anemia is a recessive trait in humans. In a cross between a father who has sickle cell anemia and a mother who is heterozygous for the sickle cell allele, what is the probability that all of their first three children will be unaffected?

a. 1/4

b. 1/2

c. None

d. 1/8

e. 1/16

d

19. Huntington disease is caused by expansion of the trinucleotide repeat region of the *HD* gene that results in the production of a Huntingtin protein with an expanded number of glutamines. An animal model with most features of this syndrome could be created by

    a. adding a transgene containing a disease-causing mutant allele of the *HD* gene to a mouse genome.

    b. adding a transgene containing a wild-type copy of the *HD* gene to a mouse genome.

    c. knocking out both copies of the *HD* gene from a mouse genome.

    d. knocking out one copy of the *HD* gene from a mouse genome.

    a

Part III: Short answer. Each question is worth 2 points, unless otherwise specified.

1. Indicate one of the four major classes of rearrangements that is most likely to be associated with each phenomenon below. There may be more than one answer. In each case, briefly explain why and whether or not the rearrangement could be caused by aberrant crossing over. Finally, explain how you could use high-throughput sequencing to detect the rearrangement. You do not have to go into the details of how to make sequencing libraries or how high-throughput sequencing works. The answer assumes that your reads are sequenced and already aligned to the reference genome.

   You may have duplicate answers (i.e., the same rearrangement could apply to multiple phenomenon below.). Assume the organism is diploid and rearranged chromosomes are rare such that the organism is heterozygous for the rearrangement.

   a. Semi-sterility

Type of rearrangement that is most likely?

Inversions or translocations.

Why?

Crossovers with inversion and translocation heterozygotes result in unbalanced gene copies.

Yes or no. Could be caused by aberrant crossing over?

Yes

How could you detect this rearrangement with high-throughput sequencing (you can draw a diagram, if helpful)?

Most easily detected with paired-end sequencing. Paired-reads would be expected to be at the expected orientation and distance when aligned to the reference genome. If there is an inversion, you would expect reads are not in the correct orientation and possibly closer or farther away than expected. If there is a translocation, the paired-reads would likely map to different chromosomes. You would not expect any difference in read coverage, unless the reads cross the breakpoints of the translocations and inversions because the breakpoint sequence would be difficult to map to. Other possible answers may be accepted.

b. Loss-of-function

Type of rearrangement that is most likely?

Deletion. Other rearrangements can be acceptable if explained correctly.

Why?

A loss of copy of a gene results in lowered expression so this is most likely to result in decreased gene activity.

Yes or no. Could be caused by aberrant crossing over?

Yes

How could you detect this rearrangement with high-throughput sequencing (you can draw a diagram, if helpful)?

You would see reduced read coverage for deletions. You could also have read pairs that map closer in the reference than expected because the deletion occurs in between. You could also see a split read that crosses the deletion. Other possible answers could be accepted.

c. Gain-of-function

Type of rearrangement that is most likely?

Duplication. Other rearrangements can be acceptable if explained correctly.

Why?

Increased copies of a gene results in increased expression so this is most likely to result in increased gene activity.

Yes or no. Could be caused by aberrant crossing over?

Yes

How could you detect this rearrangement with high-throughput sequencing (you can draw a diagram, if helpful)?

You would see increased read coverage.

You may also see paired-end reads that map in an incorrect orientation because one sequence would likely occur after another sequence, when it shouldn't in the reference. Example:

Reference	Duplication
A – <u>B – C</u> – D	A – B – C – B – C – D

You would see reads paired reads where B goes after C in the duplication, which is different than the reference.

You may also see a split read at the breakpoint of the duplication.

2. The following sequence of DNA from the genome contains a hypothetical, very short protein. Only one strand of the DNA sequence is given.

5' –AATAGCGG**ATG**TCCCCTATCGAAGGTCTTCTA**TGA**GGCCCCCTCACTAAAGCTTG– 3'

a. What is the full amino acid sequence of the short polypeptide?

MET-SER-PRO-ILE-GLU-GLY-LEU-LEU

b. You would like to use CRISPR/Cas9 to create a knockout of this polypeptide to investigate its function. This requires the design of an sgRNA. The crRNA portion of the sgRNA is typically 20nt long; however, for the purposes of this question, just indicate the first 6nt of the sgRNA sequence. Give one possible 6nt sequence that would be used to potentially knockout this gene?

Multiple answers possible. Must consider NGG sequence and must cut within the gene sequence.

c. After your targeted cut, DNA repair through NHEJ resulted in a mutated DNA sequence that caused a truncated protein. What type of protein coding mutation, most likely occurred: nonsense, missense, silent, or frameshift mutation.

Most likely it was a frameshift mutation, but nonsense mutation is also an accepted answer.

3. In a study led by scientists at the University of Chicago (Forrest et al. Cell Stem Cell 2017), they used epigenomic techniques to try to understand the function of GWAS-implicated schizophrenia risk variants. Previous GWAS studies to identify SNPs associated with schizophrenia found many in non-coding regions of the genome. Because chromatin accessibility strongly influences transcriptional activity, they hypothesized that cell-type-specific chromatin accessibility might highlight disease-relevant noncoding sequences.

a. They performed an epigenomic assay to identify accessible chromatin regions in human induced pluripotent stem cells and 27, 33, and 41 days after they were differentiated into neuronal cells. What epigenomic assay did they perform?

ATAC-Seq

b. They decided to focus on a GWAS SNP rs1198588 that is ~37 kb upstream of the transcriptional start site of a non-coding microRNA gene, *MIR137*. A separate study also examined rs1198588 and found that the schizophrenia risk allele was associated with lower expression of *MIR137* in human tissue. The scientists decided to use this information to genetically manipulate neuronal cells to see if there are any cellular phenotypes that are consistent with neuronal defects, thus modeling the

schizophrenia disease. Indicate whether each of the following neuronal cell models would be useful for their studies. Also briefly explain why or why not. You do not have to go into specific details about how to generate these cell models.

Other answers may be accepted

i. Transgenic neuronal cell overexpressing a normal human protein (2 points)

   Useful? Yes or No (circle one)

   Why or why not?
   The risk allele is not associated with expression of a non-coding gene and not a protein coding-gene. Possibly, if you had predicted targets of the miRNA gene that are protein coding, than you would predict that loss of miRNA would cause overexpression of the target protein-coding gene, so this answer may be accepted.

ii. Transgenic neuronal cell expressing normal amounts of a mutant human protein (2 points)

   Useful? Yes or No (circle one)

   Why or why not?
   The risk allele is not associated with expression of a non-coding gene and not a protein coding-gene.

iii. Transgenic neuronal cell overexpressing a normal non-coding RNA (2 points)

   Useful? Yes or No (circle one)

   Why or why not?
   The risk allele was associated with decreased expression of the miRNA, so overexpression would not be the right model.

iv. A knockout neuronal cell line (2 points)

   Useful? Yes or No (circle one)

   Why or why not?
   You could knockout the miRNA gene to mimic decreased expression.

v. A knockin neuronal cell line where the non-risk allele is replaced with the risk allele (2 points)

Useful? Yes or No (circle one)

Why or why not?

This would create the exact genetic change to the risk allele that is associated with schizophrenia. You can also test if the genetic change causes decreased expression of *MIR137.*

c. Out of the cell line models given above (i-v), which would be the best model to create.

v. A knockin neuronal cell line where the non-risk allele is replaced with the risk allele

d. The scientists looked at the phenotypes of their cell line model and saw a significant effect on neurodevelopmental phenotypes like dendritic arborization and synaptic protein accumulation in dendritic protrusions. Did these scientists find the SNP that causes schizophrenia? Why is it difficult to know what the causal SNP is? You only need to give one possible reason.

There are multiple reasons that could be given. With complex diseases, there are usually multiple variants involved. There may also be other types of complex variants like duplication or deletion that contributes, but are not typically examined in GWAS studies. They are also doing these experiments in cell lines and cannot perform genetic experiments in a whole human organism.

Initials:_____

Part IV: Extended calculations (show pedigrees and genotypes, punnett squares, etc.) YOU MUST SHOW YOUR WORK FOR FULL OR PARTIAL CREDIT

1. (**15 points**) A wildflower native to California, the dwarf lupin (Lupinus nanus) normally bears blue flowers. Occasionally, plants with pink flowers are observed in wild populations. Flower color is controlled at a single locus, with the pink allele completely recessive to the blue allele. Harding (1970) censused several lupin populations in the California Coast Ranges. In one population of lupins at Spanish Flat, California, he found 25 pink flowers and 3291 blue flowers, for a total of 3316 flowers.

   Calculate the expected allele frequencies and genotype frequencies (homozygotes and heterozygotes) if the population were in Hardy-Weinberg equilibrium.

   Let B be the blue allele and b be the pink allele, so that:
   p = frequency (B) and q = frequency (b).

   The frequency of the bb genotype = 25/3316 = q2 ,
   so q = √(0.00754) = 0.0868.

   p = 1 - q, so p = 0.913

   freq (BB) = p2 = 0.834

   freq (Bb) = 2pq = 0.158

2. (**15 points**) A married couple went to see a genetic counselor because each had a sibling with cystic fibrosis. CF is a recessive disease and neither member of the couple nor any of their four parents are affected.

   a. Draw the pedigree for this family and indicate any known genotypes.

   b. What are the chances that their child will be affected with CF?

   The chance that either parent is a carrier is 2/3. Thus the chances of having a child with genotype cf/cf is 2/3 X 2/3 X 1/4 = 4/36 = 1/9.

   c. What is the probability that their child will be a carrier of the CF mutation, but not affected?

   The child would be a carrier if its genotype were Cf/cf (or cf/+).

   This could occur if both parents were heterozygotes:
   2/3 X 2/3 X 1/2 = 4/18 = 2/9
   Or if one parent is heterozygous:
   2/3 X 1/3 X 1/2 = 2/18 = 1/9
   +
   1/3 X 2/3 X 1/2 = 1/9 = 2/9

   Thus the probability is 2/9 + 2/9 = 4/9.

3. **(30 points)** You would like to construct a mouse model for the recessive human genetic condition cystic fibrosis which is caused by a loss-of-function allele. Describe your experiment to make a good mouse model for cystic fibrosis. **Diagram any construct you would use to set up your experiment.** There are many ways to perform this experiment, so be detailed as possible.

For your mouse experiment please state:
i) what DNA construct, RNA construct, and/or protein you would introduce into the mouse cells. You may draw pictures. Indicate any selectable markers on DNA or RNA that you may introduce.
ii) whether you would put the DNA into a zygote or ES cells
iii) if introducing DNA, where in the mouse genome does the DNA integrate
iv) whether creating the mouse should involve the generation of a chimera or not
v) which additional breeding steps you would do to make the mouse you wanted

Fastest way is to generate a knockout mouse is to use CRISPR/Cas9.

i) You need to introduce three components into zygote. (1) The sgRNA that targets cleavage in an exon of the gene and has to target next to an NGG sequence. Target cleavage is preferred at the beginning of the coding sequence. The sgRNA can be introduced in an expression vector or as a synthetically made sgRNA. (2) The Cas9 protein can also be injected as a protein or as a DNA construct with the Cas9 coding region downstream of a strong promoter

ii) Since this is CRISPR, you can inject these components into a fertilized egg (zygote).

iii) Depending on the DNA construct used (particularly linear pieces of DNA), the DNA can be incorporated at a random position in the genome.

iv) Because you are injecting the mouse at the zygote stage, it is likely you are not generating a chimera; however, the final result of the CRISPR experiment may result in different mutations on both copies of the chromosome, one copy of the chromosome, or no mutation at all.

v) You want to generate a knockout mouse that is homozygous for loss-of-function alleles. After injection, the zygotes are placed in receptive females. When born, these mice are genotyped using PCR and Sanger sequencing or restriction digest. Those that show evidence of genome editing are crossed with wild-type adult mice to create F1 progeny that are heterozygous. You can, again, genotype using PCR followed by Sanger sequencing. Finally, these F1s are crossed to generate F2 progeny that are homozygous for the loss-of-function CRISPR edited alleles.